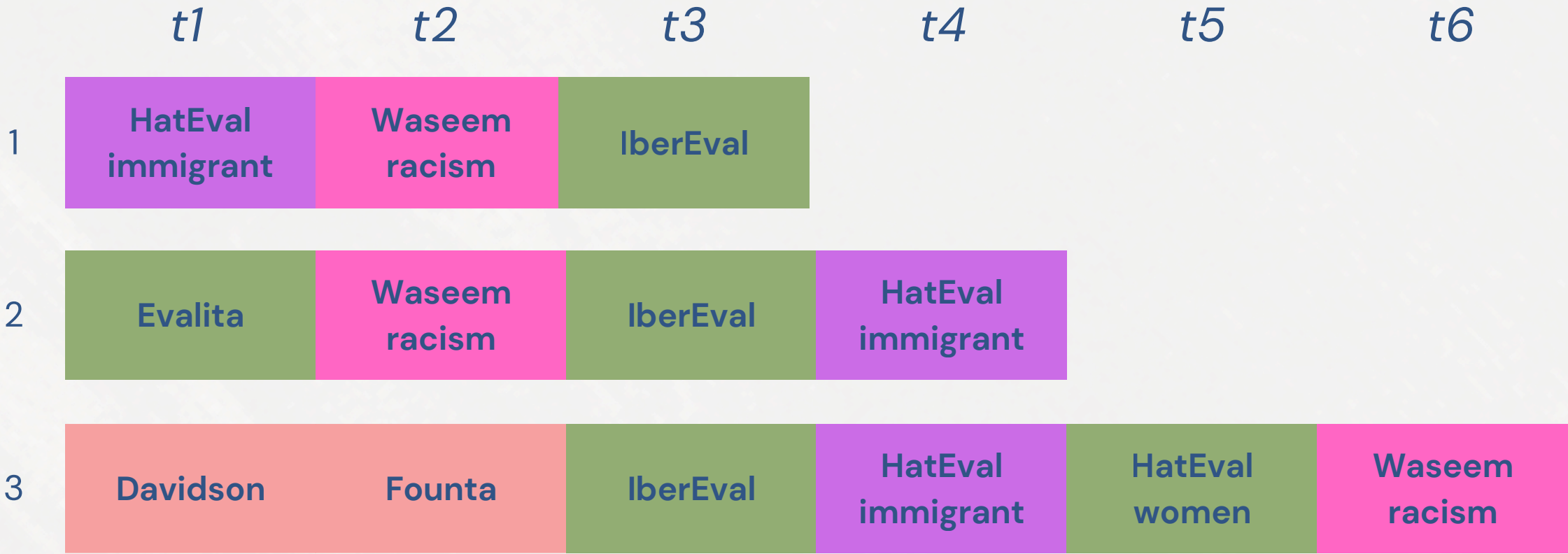


Time-Agnostic Hate Speech Detection: *a brief introduction*

Previous Work

- Continual Learning (CL) for Hate Speech Detection and **Domain Transfer**.
- PLMs + standard CL techniques.
- Results: **A-GEM** (data replay and gradient projection technique) surpassed our baselines on average, **improved Hate Class** predictions and transfer effects were more noticeable for the **longer sequences**.



PLM	CL Technique	CL Metrics		
		LAST	BWT	FWT
RoBERTa	A-GEM	0.74 $\pm$ 0.02	0.01 $\pm$ 0.05	0.33 $\pm$ 0.05
	EWC	0.66 $\pm$ 0.07	-0.10 $\pm$ 0.06	0.26 $\pm$ 0.08
	LwF	0.70 $\pm$ 0.06	-0.01 $\pm$ 0.07	0.31 $\pm$ 0.05
	MAS	0.67 $\pm$ 0.03	-0.01 $\pm$ 0.01	0.33 $\pm$ 0.05
	SEQ-FT	0.68 $\pm$ 0.07	-0.06 $\pm$ 0.08	0.27 $\pm$ 0.07
HateBERT	A-GEM	0.72 $\pm$ 0.02	0.02 $\pm$ 0.06	0.12 $\pm$ 0.04
	EWC	0.68 $\pm$ 0.04	-0.03 $\pm$ 0.05	0.09 $\pm$ 0.04
	LwF	0.68 $\pm$ 0.06	0 $\pm$ 0.06	0.14 $\pm$ 0.02
	MAS	0.63 $\pm$ 0.03	0.02 $\pm$ 0.03	0.20 $\pm$ 0.05
	SEQ-FT	0.69 $\pm$ 0.06	-0.01 $\pm$ 0.09	0.06 $\pm$ 0.02
fBERT	A-GEM	0.72 $\pm$ 0.01	0 $\pm$ 0.02	-0.07 $\pm$ 0.11
	EWC	0.68 $\pm$ 0.03	-0.01 $\pm$ 0.03	-0.11 $\pm$ 0.08
	LwF	0.69 $\pm$ 0.04	0.01 $\pm$ 0.04	-0.10 $\pm$ 0.08
	MAS	0.60 $\pm$ 0.01	0.01 $\pm$ 0.01	-0.07 $\pm$ 0.07
	SEQ-FT	0.70 $\pm$ 0.02	0 $\pm$ 0.02	-0.11 $\pm$ 0.08

Continual Learning Results Aggregated over Experiments

Time Agnostic HSD



Literature Review

Time-agnostic/aware Techniques

- **Combinations/alignment of time-specific embeddings** (Yao et al., 2018).
- **Continual Learning** and **Time-invariant techniques** for Legal Text classification (T.y.s.s et al., 2024).
- Non-CL techniques assume available data before the training starts.
- Time-invariant techniques perform **worse** than Continual Learning Techniques, specially **LoRA** and **Adapter** methods (T.y.s.s et al., 2024).

Time-related Performance Degradation and Mitigation

- Models tend to perform **worse on future data** than in previous data: performance degrades with time (Loureiro et al., 2022).
- Mitigation: train with dataset of a time **as close** to the test set as possible (Florio et al, 2020).
- Common baselines: standard finetuning for each time period individually, random chronological splits (Jin et al. 2023), concatenating splits.
- Source of errors: **including “event and topic information”** (Jin et al., 2023; Röttger and Pierrehumbert, 2021).

Literature Review

- Röttger and Pierrehumbert (2021) tackle how time adaptation affects **both upstream MLM and its effect on downstream political document classification tasks** concluding that while Temporal Adaptation (TA) is preferable to no adaptation in upstream MLM, Domain Adaptation can outperform TA when it comes to further downstream tasks.
- T.y.s.s et al. (2024) implements **incremental training paradigms** for downstream legal classification tasks. All in all, temporally invariant methods for domain adaptation perform the worst and LoRA and Adapters work the best for them. CL models outperform their standard finetuning baselines but time-invariant methods don't always reach the baseline.

Challenges for *Time-agnosticism*

1

Data is assumed to be already **pre-split** by time and it is **available**

2

When to split a data and **conclude** the time period of said data?

3

The time dimension is assumed to be known

4

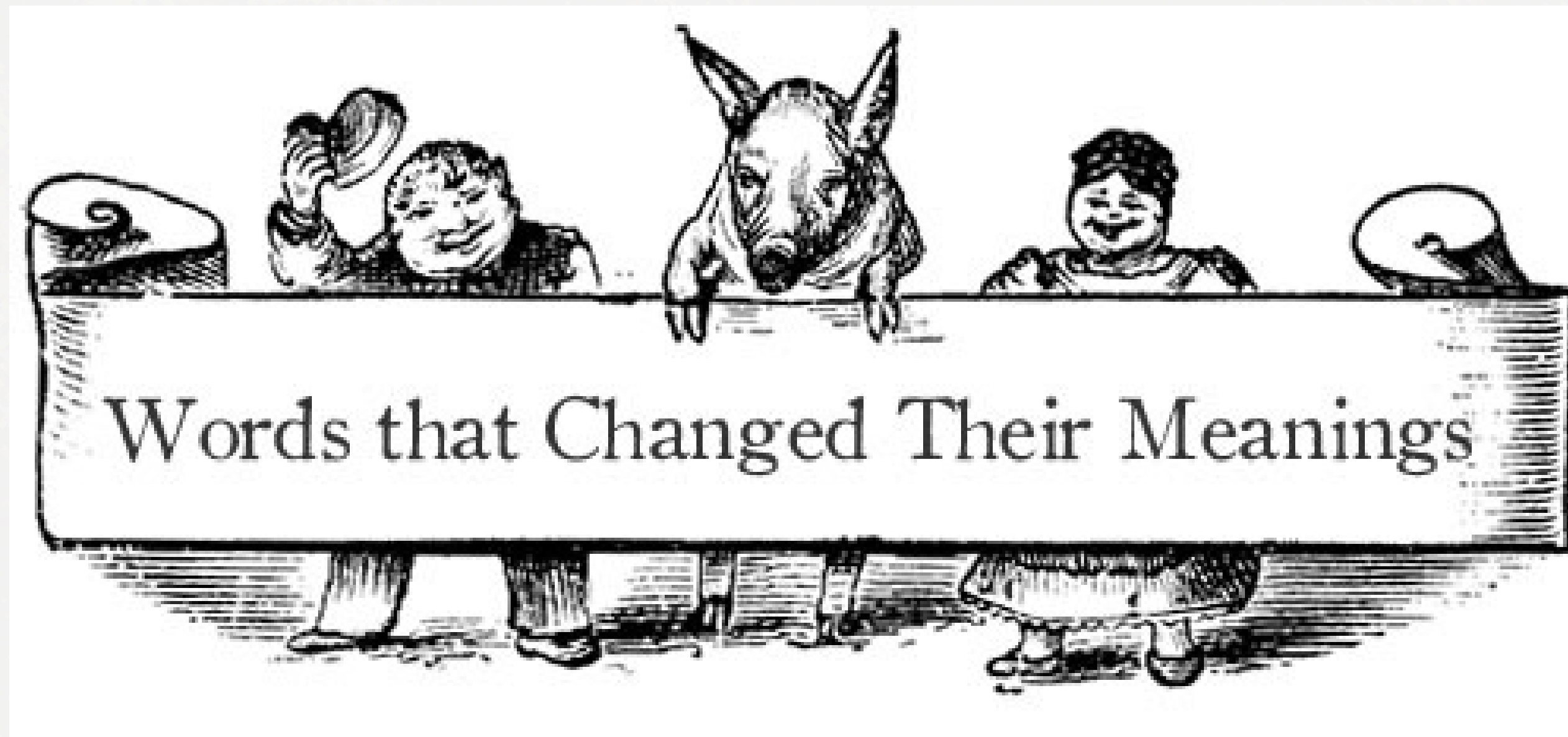
The **further** the test data is from the train data the **worse** the performance

4

Static methods ***chase the future***

Related Field: Lexical Semantic Change Detection (LSCD)

Does a word change its meaning with respect to two Corpora?



Related Field: Lexical Semantic Change Detection (LSCD)

RECENT WORKS

- Kiyama et al. (2025): compute a **similarity matrix** between timed embeddings of the same word through time, **clustering** the similarity matrices for different words to categorize words that exhibit similar behaviour.
- Wang and Choi (2023): zero shot **prompting GPT-4** for LSCD outperforms BERT in TempoWiC, which exhibited high prompt sensitivity.
- Hoeken et al. (2023): measure lexical change of **hate speech in historical data** (16th century early modern English religious writings during the protestant reform) by projecting embeddings into a “hate” dimension.

— Challenges for LSCD

1

Comparison Corpora **already split**

2

Pair-wise corpora: what words to compared?

3

Change in **use** != change in **meaning***

Possible RQs and directions?

- How do reasoning models' rationales change after time-invariant/aware finetuning? Would a model go from giving topic/event specific reasons to more general abstract ones?
- Given an already instruction-tuned LLM in hate speech detection, is there a way to continue MLM upstream training to make the LLM less sensitive to time without having to continually instruction tune it more? Benefit from unlabelled current data without losing prompting performance.
- Is there such a thing as Mixture-of-Time-Experts possible? Current PLM time-invariant embedding techniques resemble MoE architecture.
- Part of the degradation in performance comes from language evolution, would training with some kind of noise (like with Low-Resource Language Tokenizers for example), make models more time-robust?

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